

AISHWARYA HALVAGAL VEERESH

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Summary

- 4+ years of experience in data engineering, with a solid foundation in designing and optimizing ETL pipelines, data models, and cloud infrastructures.
- Proficient in data analysis and engineering using Python, SQL, AWS, and Azure, with expertise in data cleaning, data warehousing, data lakes, and data visualization through tools like Power BI and Tableau.
- Extensive experience in data analysis, including data cleaning, transformation, and integration, leveraging advanced tools and techniques to extract actionable insights from large datasets.
- Expertise in developing and deploying interactive dashboards and visualizations using Power BI and Tableau, facilitating data-driven decision-making across diverse business functions.
- Strong background in implementing CI/CD pipelines, automating data workflows with tools like Apache Airflow and Azure Data Factory, and optimizing system performance for real-time analytics.

Education

University at Buffalo, Buffalo, New York

Aug 2022 - Dec 2023

Master of Science in Data Science

SDM College of Engineering, Dharwad, India

Jun 2016 - Aug 2020

Bachelor of Engineering in Computer Science and Engineering

Technical Skills

Languages and Tools: SQL, Python, R, AWS (S3, Redshift, Lambda, Glue, QuickSight), Informatica, DAX, Spark, Google Cloud, dbt.

BI Tools/Database: Google BigQuery, Power BI, Tableau, SSIS, SSAS, SSRS, MySQL, PostgreSQL, Oracle, Snowflake, Excel, Mongo DB, Teradata, Azure Data Factory.

Libraries and Technologies: Pandas, NumPy, Scikit-learn, SciPy, Matplotlib, GitHub, Git, Databricks, Apache Spark, Azure Data Factory, Jupyter Notebook, Agile/Scrum, Waterfall, Spark, Databricks, Apache Kafka, Apache Airflow, CI/CD Pipelines.

Certifications: Azure Fundamentals (AZ 900), Hacker Rank SQL Advanced Certification.

Experience

Third Estate Ventures, Buffalo, New York

Data Engineer

Jan 2024 – Present

- Developed a scalable ETL workflow using Python, integrating data from 5+ sources into AWS Redshift, which cut data error rates by 50% and improved data preparation efficiency by 3X.
- Built and optimized data models in AWS Redshift, handling terabytes of data from user interactions, leading to a 30% reduction in query execution times and faster data retrieval.
- Orchestrated AWS S3 bucket creation and management with stringent IAM roles policies to reinforce data security and access control.
- Developed automated workflows using Airflow and Python for daily incremental data loads, enabling seamless data transition from traditional RDBMS to Azure Data Lake, optimizing data availability and reliability.
- Developed and managed data lake solutions using Delta Lake on Azure Databricks, ensuring high data quality, consistency, and accessibility for advanced analytics.
- Spearheaded the Continuous Integration and Continuous Deployment (CI/CD) of applications into Azure Cloud, enhancing operational agility and system reliability.
- Crafted and deployed JSON scripts in Azure Data Factory, leveraging SQL Activity for efficient data processing and integration, highlighting technical proficiency and advanced deployment capabilities.
- Designed ETL framework in PySpark for daily data processing, including error handling and logging, resulting in increased efficiency.

Merchants Insurance Group, Buffalo, New York

Business Intelligence Intern

Jun 2023 – Aug 2023

- Defined agile sprint requirements and delivered comprehensive reporting data, detailed insights, and interactive dashboards to stakeholders, facilitating informed decision-making and strategic planning.
- Designed and developed interactive, user-friendly dashboards and visualizations in Tableau, incorporating advanced charting techniques, calculations, and data blending to present complex data in an intuitive manner.

- Gathered and documented detailed business requirements, translating them into technical specifications and design documents to ensure clear communication and alignment between business and technical teams.
- Partnered with data engineers and stakeholders to refine data mapping rules and transformation logic, ensuring a seamless transition of complex CSV data into the client's structured database and maintaining data integrity.
- Minimized infrastructure costs by closely monitoring resource utilization with AWS CloudWatch, implementing effective optimizations that enhanced system performance and scalability while reducing operational expenses.
- Created a robust data extraction framework using Python scripts to extract raw data from over 100 XML files, identifying and addressing anomalies and inconsistencies in the collated data to ensure accuracy and reliability.

KPMG, India

Data Engineer

Jan 2020 – Jul 2022

- Developed robust data pipelines for the Customer Insights and Personalization Platform using Python and SQL, optimizing ETL processes to handle and transform large-scale customer datasets for personalized recommendations.
- Administered and optimized MS SQL Server databases by designing efficient schemas and composing complex queries, ensuring seamless data storage and retrieval to support real-time insights for over 1 million customers.
- Applied Matplotlib libraries for advanced data manipulation and analysis, facilitating the extraction of actionable insights from customer behavior data to drive personalization strategies.
- Leveraged AWS services, including S3, EC2, Lambda, and CloudWatch, to construct and deploy scalable cloud-based infrastructure, enhancing the platform's performance and reliability for processing customer data in real time.
- Utilized Big Data tools such as Hadoop to process and analyze extensive volumes of customer data, significantly improving the speed and accuracy of insights generation for targeted marketing campaigns.
- Developed interactive dashboards and detailed reports using Power BI and MS Excel, presenting key customer metrics and trends to stakeholders, enabling data-driven decisions for enhancing engagement across 10+ marketing initiatives.

Data Analyst

- Engineered robust data pipelines using Python and Pandas to conduct comprehensive data mining and cleansing operations for website analytics data, significantly improving data accuracy for critical business analyses across multiple departments.
- Designed complex SQL queries in Oracle databases, extracting and transforming millions of website interaction records monthly, providing actionable insights to enhance user experience and inform strategic decision-making.
- Developed and maintained over 15 interactive Power BI dashboards to visualize key performance indicators (KPIs) such as page views, bounce rates, and conversion rates, driving data-driven decision-making among senior management.
- Implemented version control best practices using GitHub, managing a repository of over 50 analytical scripts focused on website analytics, ensuring efficient collaboration and code integrity among the data team.
- Conducted advanced statistical analyses and predictive modeling techniques to uncover hidden patterns and correlations in user behavior, leading to significant improvements in user engagement and conversion rates.
- Streamlined data collection processes by automating report generation with Python and MS Excel macros, reducing manual data entry by 20 hours per month.

Beyond Key Solutions, India

Data Analyst Intern

Jul 2019 – Dec 2019

- Leveraged Python to extract, transform, and analyze sales data from multiple sources, developing automated scripts that reduced data processing time by 50 hours per month and improved the overall efficiency of the sales data analysis workflow.
- Designed and optimized MySQL databases to store and manage large volumes of sales data, implementing indexing strategies and query optimizations that enhanced data retrieval speed.
- Utilized Power BI and MS Excel to create comprehensive, interactive dashboards and reports, translating intricate sales data patterns into visually compelling presentations that drove strategic decision-making across sales departments.
- Conducted data cleansing and quality assurance on sales data, applying statistical methods and custom algorithms to rectify anomalies, resulting in a 25% increase in data accuracy and reliability for critical sales reports.
- Performed advanced statistical analyses on sales data using NumPy and other Python libraries, including regression modeling, hypothesis testing, and time series analysis, to uncover key business insights that contributed to operational efficiency in sales processes.

Projects

IPL Data Analysis (Tools/Tech: AWS S3, Databricks, Spark, SQL Analytics, Tableau)

- Orchestrated data collection into AWS S3, followed by preprocessing and analysis using Databricks and Spark. Utilized SQL Analytics for in-depth querying and created Tableau dashboards to visualize key insights on player performance, team dynamics, and match outcomes.